Resistor Values (□□□□)

R1	10kΩ	±1%
R2	2.2kΩ	±5%
R3	100Ω	±1%
R4	1MΩ	±5%
R5	4.7kΩ	±1%
R6	47kΩ	±1%
R7	220Ω	±5%
R8	330Ω	±1%
R9	10Ω	±5%
R10	0Ω	(jumper)
R11	1.5kΩ	±1%
R12	6.8kΩ	±1%
R13	22kΩ	±5%
R14	100kΩ	±1%
R15	470Ω	±5%
R16	3.3kΩ	±1%
R17	75kΩ	±1%
R18	150Ω	±5%
R19	2.7kΩ	±1%
R20	5.1kΩ	±1%

Capacitor Values (□□□□)

C1	10µF	16V	Electrolytic
C2	100µF	25V	Electrolytic
C3	100nF	50V	Ceramic
C4	0.1µF	50V	Ceramic
C5	22µF	10V	Tantalum
C6	1µF	25V	Ceramic X7R
C7	4.7µF	16V	MLCC
C8	10nF	100V	Ceramic
C9	47µF	35V	Electrolytic
C10	220µF	25V	Electrolytic
C11	100pF	50V	NP0/C0G
C12	2.2µF	10V	Ceramic X5R
C13	470µF	16V	Electrolytic
C14	33pF	50V	NP0
C15	1000µF	6.3V	Electrolytic
C16	0.01µF	50V	Ceramic

Decoupling: 100nF near each power pin
Bulk: 10µF + 0.1µF per rail

IC Pin Assignments (□□□□□□)

U1: ESP32-WROOM-32E

1	EN	14	GPIO12
2	GPIO36	15	GPIO14
3	GPIO39	16	GPIO27
4	GPIO34	17	GPIO26
5	GPIO35	18	GPIO25
6	GPIO32	19	GPIO33
7	GPIO33	20	GPIO32
8	GPIO25	21	GPIO35
9	GPIO26	22	GPIO34
10	GPIO27	23	GPIO39
11	GPIO14	24	GPIO36
12	GPIO12	25	GND
13	GND	26	VDD 3.3V

U2: STM32F103C8T6 (LQFP-48)

1	VBAT	13	PA3/TIM2_CH4
2	PC13	14	PA4/SPI1_NSS
3	PC14-OSC	15	PA5/SPI1_SCK
4	PC15-OSC	16	PA6/SPI1_MISO
5	PD0-OSC	17	PA7/SPI1_MOSI
6	PD1-OSC	18	PB0/TIM3_CH3
7	NRST	19	PB1/TIM3_CH4
8	VSSA	20	PB10/I2C2_SCL
9	VDDA	21	PB11/I2C2_SDA
10	PA0-WKUP	22	VSS_2
11	PA1	23	VDD_2
12	PA2	24	BOOT0

Power Supply & Voltage Labels (□□□□)

Net Label	Voltage	Current	Description
VCC	5.0V	500mA	Main supply
VDD	3.3V	300mA	Digital core
VDDA	3.3V	50mA	Analog supply
VREF	2.5V	10mA	ADC reference
VBUS	5.0V	1000mA	USB bus power
VBAT	3.7V	200mA	Battery input
VSYS	3.7-5.5V	2000mA	System input
VIN	7-12V	2000mA	External input
VOUT	3.3V/5V	1000mA	Regulator out
+5V	5.0V	2000mA	Positive rail
+3.3V	3.3V	1500mA	Logic supply
+12V	12.0V	3000mA	Motor/relay
+24V	24.0V	5000mA	Industrial
-5V	-5.0V	100mA	Op-amp neg
-12V	-12.0V	300mA	RS-232/comms
GND	Ground	—	Digital ground
AGND	Analog GND	—	Analog ground
PGND	Power GND	—	Power ground
EARTH	Chassis	—	Safety earth

Connector Pinouts (□□□□□)

J1: USB-C Receptacle (USB 2.0)

A1	GND	B1	GND
A2	SSTXp1	B2	SSRXp1
A3	SSTXn1	B3	SSRXn1
A4	VBUS	B4	VBUS
A5	CC1	B5	CC2
A6	D+	B6	D+
A7	D-	B7	D-
A8	SBU1	B8	SBU2
A9	VBUS	B9	VBUS
A10	SSRXn2	B10	SSTXn2
A11	SSRXp2	B11	SSTXp2
A12	GND	B12	GND

J2: 2.54mm Pin Header (1x8)

1	VCC 5V
2	GND
3	SDA (I2C Data)
4	SCL (I2C Clock)
5	GPIO4 / PWM
6	GPIO5 / ADC_IN
7	UART TX
8	UART RX

Bill of Materials — Part 1 (□□□□)

Ref	Value	Package	Qty	Note
R1,R2,R3	10kΩ ±1%	0805 SMD	3	Pull-up resistors
R4,R5	2.2kΩ ±5%	0805 SMD	2	LED current limit
R6	0.1Ω 1W	2512 SMD	1	Current sense shunt
C1,C2,C3	100nF 50V	0603 X7R	3	Decoupling caps
C4,C5	10µF 16V	0805 X5R	2	Bulk capacitance
C6	22µF 25V	TH 5mm	1	Input filter
C7,C8	22pF 50V	0603 NP0	2	Crystal load caps
U1	ESP32-WROOM	Module	1	MCU module
U2	AMS1117-3.3	SOT-223	1	3.3V LDO regulator
U3	CH340C	SOP-16	1	USB-UART bridge
U4	INA219	SOT-23-8	1	I2C current sensor
D1,D2	1N4148	SOD-123	2	Signal diodes
D3	SS34	SMA	1	Schottky 3A/40V
LED1	Green	0805 SMD	1	Power indicator
LED2	Blue	0805 SMD	1	Status LED
Q1	2N7002	SOT-23	1	N-channel MOSFET
Q2	SI2301	SOT-23	1	P-channel MOSFET
L1	10µH 1A	CD54	1	Buck inductor
X1	32.768kHz	3.2x1.5mm	1	RTC crystal
Y1	40MHz	5x3.2mm	1	Main oscillator

Bill of Materials — Part 2 (□□□□□)

Ref	Value	Package	Qty	Note
J1	USB-C 16P	SMD	1	USB connector
J2	2.54mm 1x8P	TH	1	GPIO header
J3	JST-XH 2P	TH	1	Battery connector
J4	SMA-KE	Edge	1	Antenna connector
SW1	Tactile 6mm	TH	1	Reset button
SW2	Tactile 6mm	TH	1	Boot/User button
F1	500mA PTC	1206 SMD	1	Resettable fuse
RV1	TVS 5V	SOD-323	1	ESD protection
T1	1:1 10BaseT	SMD-16	1	Ethernet transformer
P1	2.54mm 2x3P	TH	1	SWD debug header
P2	2.54mm 2x4P	TH	1	SPI expansion
HS1	Heatsink 14mm	Al	1	LDO heatsink
TP1,TP2	Test Point	SMD	2	GND reference
JP1	2.54mm 1x3P	TH	1	Power select jumper
BZ1	Piezo 5V	TH 12mm	1	Buzzer
CN1	FFC 10P 0.5mm	SMD	1	LCD flex cable

Net Labels & Signal Names (□□□□)

Analog Signals:

```
AIN0, AIN1, AIN2, AIN3          ADC input channels
VREF+                            ADC positive reference
VREF-                            ADC negative reference
TEMP_SENSOR                      NTC temperature probe
CURRENT_SENSE                    INA219 shunt voltage
```

Digital I/O:

```
GPIO0 ~ GPIO39                   ESP32 GPIO pins
UART0_TX, UART0_RX               Serial console (CH340)
UART1_TX, UART1_RX               RS-485 communication
I2C0_SCL, I2C0_SDA               Sensor bus (100kHz)
SPI_MOSI, SPI_MISO, SPI_SCK      SPI Flash/LCD bus
SPI_CS0, SPI_CS1                 Chip select lines
PWM1_OUT, PWM2_OUT               LED dimming output
ENCODER_A, ENCODER_B             Rotary encoder inputs
INT_ACCEL                        Accelerometer interrupt
```

Power Nets:

```
VBUS, VSYS, VBAT, VDD_3V3, VDD_1V8
VDD_CORE, VDD_IO, VDD_ANA, VDD_RF
GND, PGND, AGND, DGND, GND_PLANE
```

Diodes & Transistors (□□□□□□□)

D1	1N4148	Signal diode, 100V/200mA
D2	1N4007	Rectifier, 1000V/1A
D3	SS34	Schottky, 40V/3A, SMA
D4	BZX84C3V3	Zener 3.3V, 300mW, SOT-23
D5	BZX84C5V1	Zener 5.1V, 300mW, SOT-23
D6	1N5819	Schottky, 40V/1A, DO-41
D7	MBR0520L	Schottky, 20V/500mA, SOD-123
D8	LED_RED	Red LED, 2.0Vf, 20mA, 0805
D9	LED_GREEN	Green LED, 2.2Vf, 20mA, 0805
D10	LED_BLUE	Blue LED, 3.3Vf, 20mA, 0805

Q1	2N2222A	NPN BJT, 40V/800mA, TO-92
Q2	2N3904	NPN BJT, 40V/200mA, SOT-23
Q3	2N7002	N-MOSFET, 60V/300mA, SOT-23
Q4	AO3400A	N-MOSFET, 30V/5.7A, SOT-23
Q5	SI2301	P-MOSFET, -20V/-2.3A, SOT-23
Q6	BSS138	N-MOSFET level shifter
Q7	BC547B	NPN BJT, 45V/100mA, TO-92
Q8	BC557B	PNP BJT, -45V/-100mA, TO-92
Q9	IRFZ44N	N-MOSFET, 55V/49A, TO-220
Q10	IRF9540N	P-MOSFET, -100V/-23A, TO-220

IC Part Numbers & Markings (□□□□)

U1	ESP32-WROOM-32E-N4	Espressif MCU
U2	STM32F103C8T6	ARM Cortex-M3
U3	ATmega328P-AU	AVR 8-bit MCU
U4	CH340C	USB-UART bridge
U5	CP2102N-A02-GQFN28	USB-UART bridge
U6	AMS1117-3.3	LDO 3.3V/1A
U7	MP1584EN	Buck converter
U8	XL1509-3.3	Buck 3.3V/2A
U9	TPS54331	Buck 3A TI
U10	MAX232ESE	RS-232 driver
U11	MAX485ESA	RS-485 transceiver
U12	SN65HVD230	CAN transceiver
U13	INA219BIDGSR	I2C current sensor
U14	BME280	Temp/Hum/Press
U15	MPU6050	IMU 6-axis
U16	W25Q128JVSIQ	SPI Flash 128Mb
U17	PCA9685PW	PWM driver 16ch
U18	TCA9548APWR	I2C mux 8ch
U19	MCP23017-E/SP	I2C GPIO expander
U20	ADS1115IDGSR	I2C ADC 16-bit

Component Reference Designators (□□□□)

R1 R2 R3 R4 R5 R6 R7 R8 R9 R10

R11 R12 R13 R14 R15 R16 R17 R18 R19 R20

R21 R22 R23 R24 R25 R26 R27 R28 R29 R30

C1 C2 C3 C4 C5 C6 C7 C8 C9 C10

C11 C12 C13 C14 C15 C16 C17 C18 C19 C20

U1 U2 U3 U4 U5 U6 U7 U8 U9 U10

U11 U12 U13 U14 U15 U16 U17 U18 U19 U20

J1 J2 J3 J4 J5 J6 J7 J8

D1 D2 D3 D4 D5 D6 D7 D8 D9 D10

L1 L2 L3 L4 L5 L6

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8

Y1 Y2 X1 X2

LED1 LED2 LED3 LED4 LED5

SW1 SW2 SW3 SW4

TP1 TP2 TP3 TP4 TP5 TP6

F1 F2 RV1 RV2

P1 P2 P3 P4

JP1 JP2 JP3

CN1 CN2 CN3

BZ1 HS1

Schematic Text Blocks (□□□□□□)

```
POWER INPUT AND REGULATOR
+5V     +5V     +5V     +3.3V
U2:  NCP1117-ADJ
  1 ADJ    2 VO    3 VI
C1:  10µF     C2:  100nF     C3:  10µF
```

```
Buck Converter for VSYS 5V
D3:  Green LED
L1:  SWPA4030S3R3MT  (3.3µH)
  1  IN    2  OUT
+5V regulated output
```

```
Microcontroller Section
U1:  pi_pico (Raspberry Pi Pico)
  1 GP0           8  GND
  2 GP1           9  GP6
  3 GND          10  GP7
  4 GP2          11  GP8
  5 GP3          12  GP9
  6 GP4          13  GND
  7 GP5          14  GP10
Y1:  12MHz Crystal
```

Schematic Blocks (cont.)

```
USB Interface
J1:  USB_B_Micro
  1 VBUS    2 D-    3 D+    4 ID    5 GND
D1:  D_Schottky (SS14)
F1:  500mA PTC Fuse
```

```
Display Connector
J3:  Conn_01x07_Pin (SPI Display)
  1 GND          2 VCC 3.3V
  3 SCL (SCK)    4 SDA (MOSI)
  5 RES (RESET)  6 DC  (A0/Data-Cmd)
  7 CS  (Chip Sel)
```

```
Audio Output
U3:  LM386N-1  (Audio Amplifier)
  1 GAIN    2 -IN    3 +IN
  4 GND     5 VOUT   6 VS
  7 BYPASS  8 GAIN
R5:  10kΩ  (Volume pot)
SPK1:  8Ω 0.5W Speaker
```

```
RTC Section
U10:  DS3231M  (I2C RTC)
Y2:   32.768kHz Crystal
BAT1: CR2032 (Backup Battery)
```

Common Component Values (□□□□□)

Resistors (1% E96 series):
10Ω 22Ω 47Ω 100Ω 220Ω 470Ω 1kΩ
2.2kΩ 4.7kΩ 10kΩ 22kΩ 47kΩ 100kΩ
1MΩ 2.2MΩ 4.7MΩ 10MΩ

Capacitors:
10pF 22pF 33pF 100pF 1nF 10nF 100nF
0.1µF 0.22µF 0.47µF 1µF 2.2µF 4.7µF
10µF 22µF 47µF 100µF 220µF 470µF
1000µF 2200µF 4700µF

Inductors:
1µH 2.2µH 3.3µH 4.7µH 6.8µH 10µH
22µH 33µH 47µH 68µH 100µH 220µH
470µH 1mH 2.2mH 10mH

Voltages:
1.2V 1.8V 2.5V 3.0V 3.3V 5.0V
9V 12V 15V 24V 48V
-5V -12V -15V

Tolerances: ±0.1% ±0.5% ±1% ±2% ±5% ±10% ±20%
Package: 0201 0402 0603 0805 1206 1210 2512
Package: SOT-23 SOT-223 SOIC-8 TSSOP-16 QFN-32

Test Point Assignments & Silkscreen (□□□□□□)

Silkscreen Labels on PCB (PCB□□□□):

PWR	RST	BOOT	TX	RX
SDA	SCL	MOSI	MISO	SCK
3.3V	5V	GND	VIN	VOUT
A0	A1	A2	D0	D1
CH1	CH2	TRIG	ECHO	
ANT	NFC	IR	BT	WIFI
PROG	DEBUG	RESET	USER	

Test Point Assignments:

TP1	GND	Test hook (black)
TP2	+3.3V	Test pad 1mm
TP3	+5V	Test pad 1mm
TP4	UART0_TX	Test via 0.5mm
TP5	UART0_RX	Test via 0.5mm
TP6	I2C0_SCL	Test pad 1mm
TP7	I2C0_SDA	Test pad 1mm
TP8	PWM1_OUT	Test via 0.5mm
TP9	RESET_N	Test pad 1mm
TP10	VBUS_SENSE	Test via 0.5mm

Large Labels (□□□□)

10kΩ

2.2kΩ

100Ω

4.7kΩ

47kΩ

1MΩ

100nF

10µF

22µF

0.1µF

220µF

1µF

Large Labels (□□□□)

VCC 5V

VDD 3.3V

GND

VIN 12V

VOUT 3.3V

VBUS 5V

VBAT 3.7V

+12V

-12V

+24V

AGND

Large Labels (□□□□)

___

U1

U2

U3

R1

R2

C1

C2

J1

D1

L1

Q1

Y1

LED1

SW1

TP1

F1

Large Labels (□□□□)

ESP32

STM32

CH340

AMS1117

INA219

MPU6050

MAX232

CP2102

W25Q128

ATmega328P

BME280

MCP23017

PCA9685

ADS1115

SN65HVD230

Large Labels (□□□□)

SDA

SCL

MOSI

MISO

SCK

TX

RX

PWM

ADC

GPIO

I2C

SPI

UART

INT

RST

EN

CS

CLK

D+

D-